

Multiresponsive Graphene-Aerogel-Directed Phase-Change Smart Fibers

Guangyong Li, Guo Hong, Dapeng Dong, Wenhui Song, and Xueting Zhang*

Wearable devices and systems demand multifunctional units with intelligent and integrative functions. Smart fibers with response to external stimuli, such as electrical, thermal, and photonic signals, etc., as well as offering energy storage/conversion are essential units for wearable electronics, but still remain great challenges. Herein, flexible, strong, and self-cleaning graphene-aerogel composite fibers, with tunable functions of thermal conversion and storage under multistimuli, are fabricated. The fibers made from porous graphene aerogel/organic phase-change materials coated with hydrophobic fluorocarbon resin render a wide range of phase transition temperature and enthalpy ($0\text{--}186\text{ J g}^{-1}$). The strong and compliant fibers are twisted into yarn and woven into fabrics, showing a self-clean superhydrophobic surface and excellent multiple responsive properties to external stimuli (electron/photon/thermal) together with reversible energy storage and conversion. Such aerogel-directed smart fibers promise for broad applications in the next-generation of wearable systems.

Smart fibers, a fundamental unit of textiles capable of responding to various external stimuli, have become an emerging driving force for advanced flexible and wearable systems, such as sensors, actuators, bionic implants, robotics, energy harvesting devices, thermoregulating garments, and heaters.^[1] However, current textile fibers, made from both natural or semisynthetic polymers, are intrinsically electrically and thermally insulating, which limits the choice of sensors

and actuators made from the conventional electronic hardware and constrains the development of future wearable electronics and robotics. The development of nanocomposite fibers and nanofibers are currently of great interest. As reported in recent years, carbon nanostructured fibers, such as carbon nanotubes yarns and graphene fibers, promise to be lighter, stronger, tougher, and more electrical/thermal conducting in the near future.^[2] Nevertheless, more functions in response to multistimuli from complex environment and human-machine interface, such as heat, temperature, stress, light, humidity, pH etc., are still challenging.^[2a,3]

Organic phase-change materials (PCMs) and their composites have been intensively studied for the usage as heat storage and conversion from the surrounding environmental irradiation and

waste heat generated during manufacturing processes or from use of products.^[4] Conducting metal foam, carbon nanotube sponges, and graphene aerogel-encapsulated PCM composites were demonstrated substantial improvement in thermal/electrical conductivity and heat conversion efficiency compared with pure PCM, promising for applications of bulk products in the forms of sheets and panels.^[5] Some efforts were made to develop PCM-encapsulated polymeric fiber (e.g., Thermocules, Outlast) or nanofibers by utilizing both spinnability and good mechanical properties of polymers as matrix.^[6] Core-shell fiber containing TiO_2 -PVP polymer composites as the shell and PCM as the core made by coaxial electrospinning was reported to respond and buffer the environmental temperature variation by absorbing/releasing thermal energy to provide a comfortable microclimate for either human body or housing precise instruments.^[6a,7] However, such responses of the PCM-polymer fibers are difficult to collect because of the low electrical/thermal conductivity of both PCMs and polymer matrices.^[6]

Aerogel is an excellent ultralight-weight material with a 3D interconnected porous network.^[8] The extraordinary capillary force can be used to incorporate other components to obtain a novel system with multiple responsiveness and integration of more functions.^[2a,9] Mesophase-derived graphene-aerogel-based PCM composites developed in house exhibited sensitive responses to external stimuli and efficient energy conversion/storage, which is attributed to the high electrical conductivity of graphene.^[5d] However, the rigidity and uniformity of graphene aerogel-PCM composites are still beyond the requirements for robust and flexible wearable systems. It is proposed that the

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introduction of aerogel into the fiber structure should allow for impregnation of functional materials in controlled and scalable approaches, to achieve the uniformity and alignment of the structure. This maximizes the physical and mechanical properties within the fiber's confinement, and facilitates the design and manufacturing of the multiresponsive fibers.

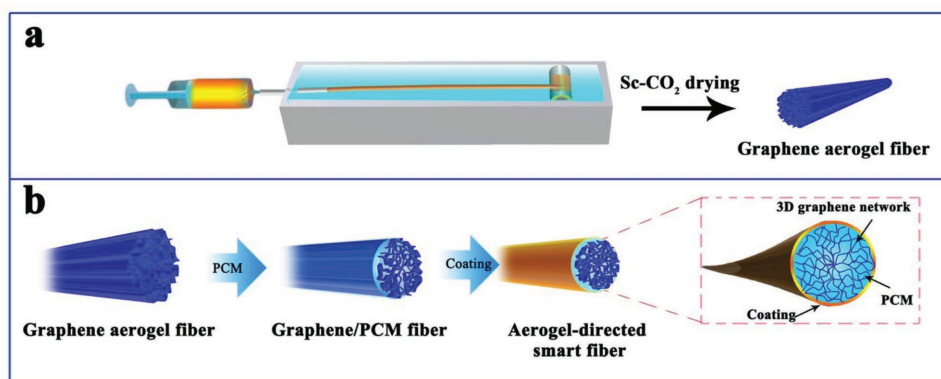
In this work, aerogel-directed smart fibers (ASFs) were fabricated by impregnating PCMs into graphene aerogel fibers (GAFs) and finished by a coating of fluorocarbon (FC) resin. The resulting smart fibers consisted of an aligned 3D graphene network within which the spacing between the graphene sheets was fulfilled by PCMs. These smart fibers exhibited tunable temperature and enthalpy of the phase transition. Both the fibers and the fabrics demonstrated superhydrophobic surfaces and excellent multiple responsive properties to external stimuli (electrical/thermal/photonic) with and without loading stresses, resulting in reversible heat conversion and storage. These tunable and multiresponsive smart fibers hold promise for applications in wearable fabrics and portable electrical devices suitable for a range of varying conditions.

The synthetic pathway of graphene/PCM ASF is shown in **Scheme 1**. It involves the synthesis and spinning of GAF, impregnation of PCMs, and then an outer coating of the hydrophobic resin. GAF was first prepared by injecting and spinning a uniform GO liquid crystal into vitamin C (VC)/HCl solution with subsequent reduction and supercritical fluid drying in sequence (Scheme 1a). The GAF is comprised of a 3D interconnected porous network with a high specific surface area ($548 \text{ m}^2 \text{ g}^{-1}$) and pore volume ($2.27 \text{ cm}^3 \text{ g}^{-1}$), (Figures S1 and S2, Supporting Information), generating the extraordinary capillary action throughout the fiber to absorb PCM resins (such as paraffin and PEG) easily. The fiber was finished by coating a protective layer of fluorocarbon resin (Scheme 1b). An optimized smart fiber consists of 83 wt% polyethylene glycol (PEG4000), a PCM-impregnated continuous graphene aerogel network, and a superhydrophobic coating. The high loading PEG with a relatively high molecular weight is envisaged to contribute to the high phase-change enthalpy and well-balanced mechanical properties of the ASF fiber among other PCM-based fibers, as showed in **Figures 1 and 2** and **Tables S1 and S2** (Supporting Information).

The field emission scanning electron microscopy (FESEM) and transmission electron microscopy (TEM) were applied to

reveal the hierarchical structure of the PEG-ASF. As shown in **Figure 1b–f** and **Figures S1 and S3** (Supporting Information), SEM images of ASF cross sections showed that the graphene sheets were stacked and wrapped in parallel, aligned with the fiber's long axis and the pores were fully filled by PEG, and a few bare graphene sheets were observed in cross section of ASF, indicating a strong adhesion between graphene and PEG. When examined at a relatively high resolution (**Figure 1c,g**), uniform nanolaminate structures between the graphene sheets and PCM resin were observed. There is a smooth and dense skin layer formed by the fluorocarbon coating around the fiber (inset of **Figure 1e** and **Figure S4** in the Supporting Information), which gives the fiber surface superhydrophobic property. Furthermore, TEM images were taken from the cross section revealed by the interface of the nanolaminate at nanoscale. The ribbon-like graphene sheets in **Figure 1h** are evenly covered by a PEG layer though still visible at the edge of PEG/graphene walls, and high-resolution TEM images (**Figure 1i–j**) shows that the nanolaminate consists of graphene sheets in a few layers and even a single layer stacking between PEG.^[10] Furthermore, the presence of PEG and FC coating on the surface of the ASFs was confirmed by XPS spectra (**Figures S5 and S6**, Supporting Information).

The resulting smart fibers consisting of the continuous PEG/graphene nanolaminates throughout the aerogel network and the superhydrophobic coating exhibited excellent strength and elasticity: it can be either tied into a knot (**Figure 1d**), entwined with commercial Outlast fiber (**Figure 2d**; **Figure S7**, Supporting Information), or woven into fabrics (**Figure 2e,g**). Mechanical measurements demonstrated that ASFs exhibited typical stress and strain profiles at room temperature, similar to polymeric fibers under tensile loading (**Figure 1k**; **Figure S8** and **Table S1**, Supporting Information). The fracture strength (12.7 MPa) of PEG-ASF is 70% higher than graphene/PEG fiber (7.5 MPa) and 775% higher than GAF (1.45 MPa), and the Young's modulus of PEG-ASF (1.2 GPa) is four times higher than graphene/PEG fiber (300 MPa) and 48 times higher than GAF (25 MPa). Clearly, PEG as a matrix and a large interface of the nanolaminates contributed most to the increase of both strength and stiffness of the fiber. The FC coating layer further enhanced both the strength and stiffness of the fiber but with a tradeoff of a reduction of the fracture strain from 2.5% to 1.52%. The overall mechanical properties



Scheme 1. Schematic description of the processes of graphene aerogel fiber (a) and graphene/PCM smart fibers (b).

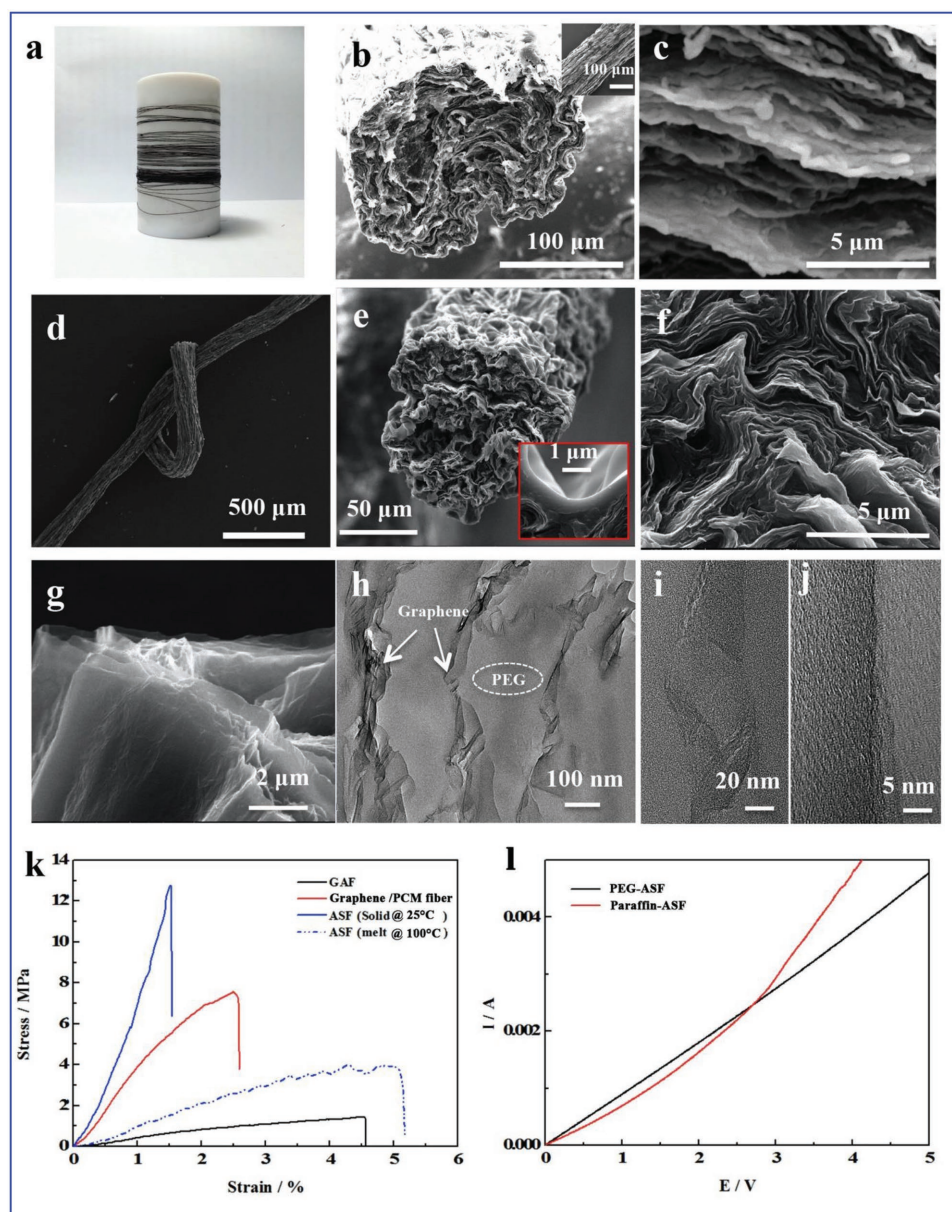


Figure 1. a) The aerogel-directed smart fiber as spun. b,c) SEM images showing the morphology of cross section of paraffin-based ASF. d–g) SEM images showing the knot and cross section of PEG-based ASF. h–j) TEM images showing a few layers of graphene coated by PEG in a PEG-based ASF. k) Tensile stress–strain curves of PEG–ASFs at 100 and 25 °C, graphene/PEG4000 composite fiber, and GAF. l) I – E curves of ASFs.

of ASF are attributed to the coherent core–shell structure of ASF: the compact stacking of graphene sheets/PEG nanolaminates and the FC coating.^[10] At the melting state of PEG (100 °C), the shape and dimension of ASF remained, while the fracture strength and Young's modulus of ASF reduced substantially compared to both ASF and composite fiber in solid (25 °C). However, the fracture strain, 5.06%, increased nearly three times than in the solid stage (1.5%), and the fracture toughness increased nearly by 50%, from 82.5 to 122.4 kJ m^{−3}, still outperforming pure GAF. This again confirms the strong interaction between graphene sheets and PEG throughout the aerogel network regardless of the softening effect of PEG in its melted state.

The electric conductivity of ASFs was investigated by I – E curves, and the ASF exhibited excellent electric conductivity with ≈ 370 S m^{−1} for PEG 83 wt% loading and ≈ 1450 S m^{−1} for PEG 53 wt% loading (Figure S9, Supporting Information), compared to the graphene aerogel fiber ($\approx 10^3$ S m^{−1})^[10] and higher than the anisotropic graphene aerogel (341 S m^{−1}).^[5d] Interestingly, ASFs with different PCM matrices exhibited different electric behaviors. As shown in Figure 1l, the ASF with PEG exhibited a linear I – E curve, while ASF with paraffin showed a less linear one with a clear turning point. This might come from different states of phase change and polarization of paraffin,^[11] which consists of a mixture of hydrocarbon molecules containing between 20 and 40 carbon atoms at different

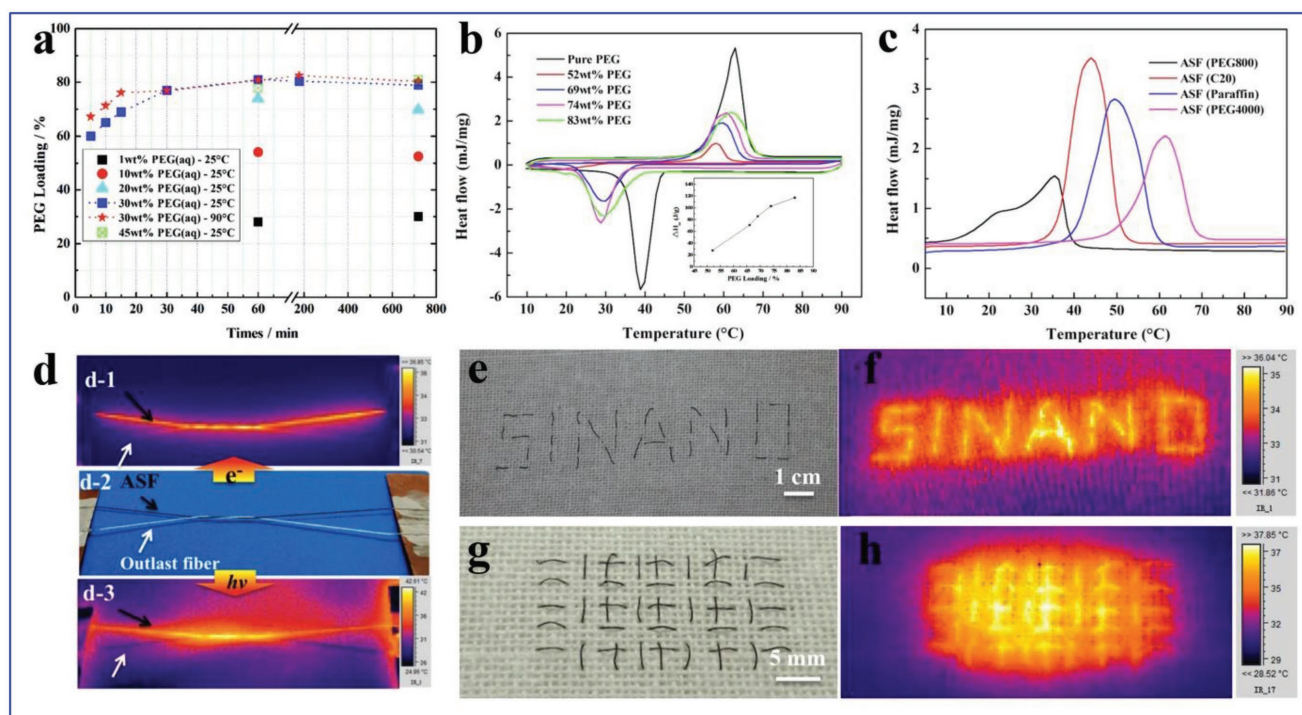


Figure 2. a) The effect of impregnation temperature and time on the loading of PEG into GAF. b) The DSC curves of ASFs and pure PEG, and the inset diagram shows the relationship between the latent heat and PEG loading content; c) the melting DSC curves of ASFs impregnated with different PCMs: PEG800, PEG4000, C20, and Paraffin. d) The infrared images of the twined PEG–ASF/Outlast fiber (d-2) under electric field (d-1) and light irradiation (d-3), respectively. e, f) The photograph (e) and IR image under light irradiation (f) of PEG–ASF in a letter pattern (SINANO) sewn on a white fabric. g, h) The photograph (g) and IR image under light irradiation (h) of ASF (black) woven together with cotton fabric (white).

voltages. We previously reported that graphene aerogel/paraffin nanocomposite generated high electroheat at high voltage.^[5d] Paraffin with short carbon chains tended to reach its phase transition earlier and polarized fast when increasing the voltage, compared to those with long carbon chains. The liquid part of paraffin could cause denser packing of graphene network of the fiber, plus more polarized paraffin at a high voltage, contributing to the nonlinear increase of the current. In contrast, the PEG used consists of uniform macromolecules with a molecular weight of $M_n = 4000$, averaged from a narrow distribution (3600–4400). The resulting small and uniform crystalline structure of PEG within the graphene network may be related to more linear I – E relationship of the PEG–ASF.

As the fundamental parameters for understanding, design, and control of the thermal energy storage and release, the thermal properties of the ASFs were investigated. The phase transition of the ASFs with different loading contents of PCM was characterized by differential scanning calorimetry (DSC), as shown in Figure 2a–c. Through changing the impregnation conditions, i.e., the concentration of PEG4000 solution, temperature, and time, the ASFs with various PEG loading from 0 to 84 wt% were produced (Figure 2a; Figures S10–S12 and Table S2 in the Supporting Information). The DSC curves in Figure 2b show a downshift of the onset point and peak temperature of each the PEG–ASF to a lower temperature in the range of 48–53 and 56–59 °C, respectively, compared to pristine PEG with a melting onset point of 57.06 °C with a peak at 64.26 °C. The less the PEG that was loaded, the lower the

transition temperature shifted. This downshift of the phase transition may be contributed by the confinement of the graphene network that reduced the PEG crystal size and crystallinity degree, and the large interface between PEG and graphene sheets that enhanced both thermal and electrical conductivity of the nanocomposite fiber (Figure 1l), and thus accelerated uniform heat dissipation within the PEG layer.^[5b,d] The other types of PCM-impregnated fibers showed a similar trend of phase change at different melting points (Figure 2c), such as PEG800 (about 24 °C), C20 (about 34 °C), and paraffin (about 42 °C).

The phase-change enthalpy is also correlated to the loading of PEG in the ASF. From Figure 2b, the pristine PEG4000 had a melting enthalpy of 188.4 J g^{−1}. In the case of ASFs with 34 wt% PEG, there was no endothermic peak scanned by DSC (Figure S10, Supporting Information) and sharp Bragg peaks in X-ray diffraction spectrum (Figure S13, Supporting Information) which indicated that the thin layer of PEG confined within the graphene sheets was in the presence of amorphous phase. When the PEG loading was increased to 52 wt% and above, the phase transition of PEG emerged and its corresponding enthalpy increased almost linearly with the loading of PEG (inset of Figure 2b). When the PEG loading reached 83 wt%, the enthalpy was measured as ≈ 124 J g^{−1}, which is much higher than that of the commercially available Outlast air-conditioning fiber (≈ 4 J g^{−1}; Figure S14, Supporting Information) and other PCMs reported elsewhere.^[6a,c,d,12] Even more intriguingly, the phase transitions of the ASF occurred in all the 20 cyclic

scans of heating and cooling at a scanning rate of $10\text{ }^{\circ}\text{C min}^{-1}$ were virtually identical (Figure S15, Supporting Information), indicating the excellent reversibility, efficiency, and stability of heat conversion and storage of the fiber.^[5d] The phase-change enthalpy was also varied by introducing other PCMs (Figure 2c), such as paraffin (about 176 J g^{-1}) and *n*-eicosane (C20, about 186 J g^{-1}). Noteworthy, the ultrathin layer of fluorocarbon coating did not interfere the phase transition of the fibers including the enthalpy, temperature of the phase transition, and thermal stability (Figures S16 and S17, Supporting Information).

The structure and properties of the nanocomposite-based ASF revealed above render unique functions in response to different external stimuli (e.g., electrical or photonic) with simultaneous change of temperature. Figure 2d shows the infrared (IR) images of a twined thread of ASF/Outlast fibers under an electric field (voltage 30 V) and light irradiation (1.0 Sun, AM 1.5, 100 mW cm^{-2} , more details are provided in Figure S18 in the Supporting Information), respectively, indicating an increase in the temperature only along the ASF rather than the Outlast fiber (Figure 2d). From the measurement of the temperature versus the irradiation time, a near equilibrium temperature of the single ASF was reached at $\approx 15\text{ s}$ (Figure S19a,b). Similar responses were also demonstrated in “SINANO” letter pattern and network woven fabrics of ASF/cotton under light irradiation (Figure 2f,h; Figures S19–S23, Supporting Information). As expected, the ASF bundle and ASF/cotton fabrics demonstrated higher heating rate and higher temperature than single ASF, indicating faster heat transfer and less heat loss within

the ASF bundle and the fabrics. The comparison between Outlast fiber, single ASF, ASF bundle, and ASF/cotton fabrics clearly indicates that the graphene network within the ASF fibers played essential multiple roles of electron transfer, photocaptor, and heat dissipation in inducing and facilitating the phase transition of PCM (PEG in this case) impregnated within the network (Figures S19–S25, Supporting Information).^[5b,d,13]

The responsive functions of ASF under different stresses and environmental conditions (temperature and moisture) were further investigated as shown in Figure 3 and Figures S19–S22 (Supporting Information). The temperature of the fiber in a bended “loop-like” shape (Figure 3a-a1) remained steady (about $40\text{ }^{\circ}\text{C}$) when the input voltage was 30 V, similar to the straight fiber at the same voltage as shown in Figure S24 in the Supporting Information, further indicating the good adhesion between the PCM and graphene fiber (Figure S26, Supporting Information). Furthermore, when the fiber was tied into a knot (Figure 3b-b1), the temperature increased to up to $72\text{ }^{\circ}\text{C}$, confirming the structural stability of fiber under 360° bending and twisting deformation.^[14] It was notified that the elevated temperature distributed mostly along the straight line shortcut of the knot instead of the whole loop, indicating a lower interface contact resistance between the self-twined part of the knot compared to the resistance along the large circular part of the knot loop. This observation may be attributed to an increase of electrical and thermal conductivity of the fiber induced by the compressed and twisted graphene networks within the knots. It is also worth mentioning that the superhydrophobic coating did

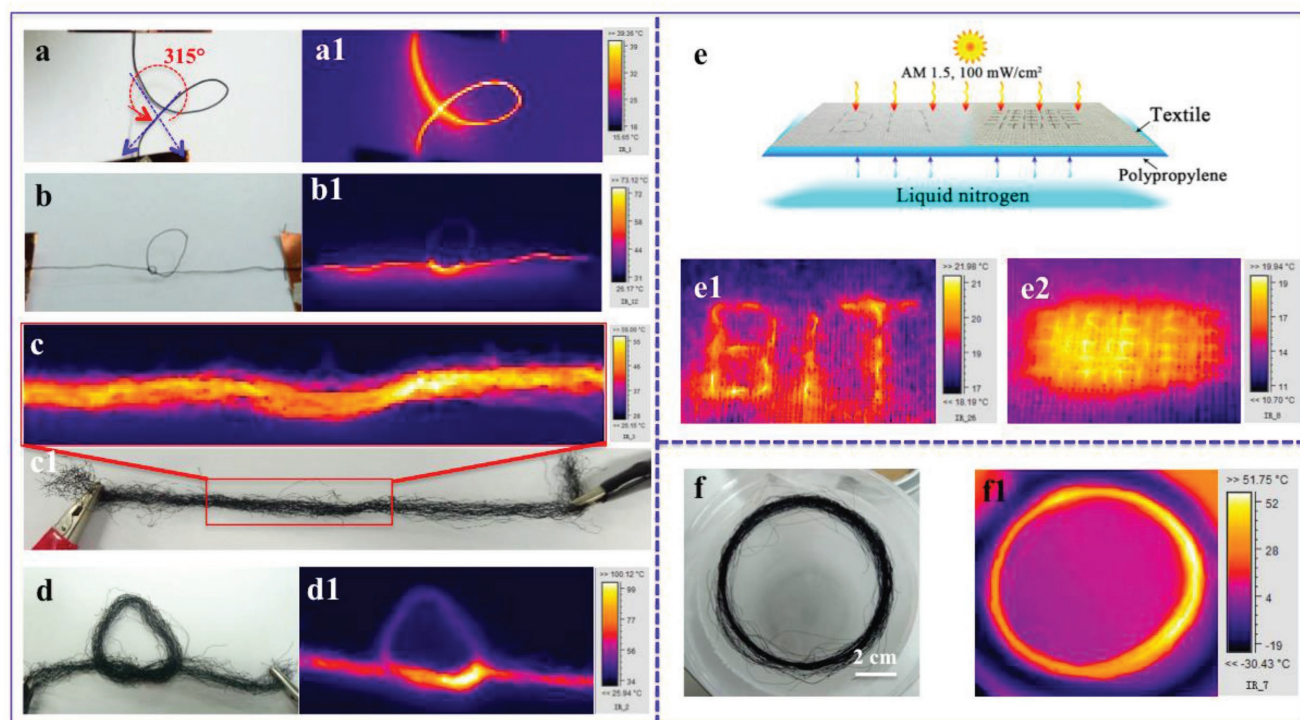


Figure 3. Photograph and IR images of the ASF by applying an input voltage about 30 V (a–d) and under solar irradiation (e–g). The single fiber, 5 cm long, was bent to a “loop-like” shape with a bending angle of 315° (no contacted site along this fiber (a–a1)) and tied into a knot (about 5 cm length (b–b1)) under 30 V electric field. The fiber bundle before (30 cm length (c–c1)) and after (d–d1) being tied into a knot under a 30 V electric field. e,f) Schematic and IR images of the photonic response of ASF textile in a cold environment (selecting a temperature of $\approx 0\text{ }^{\circ}\text{C}$). Single ASF/cotton woven fabric (e1,e2) and a fiber bundle (f–f1) placed on a cold polypropylene (PP) surface under solar illumination of 1.0 Sun.

not shade the electric conductivity; in fact, it may provide the electronic tunneling to accelerate the electron transfer between the fibers owed to high electronegativity of fluorocarbon molecules (Figure S27, Supporting Information). More pronounced electrothermal responses of a fiber bundle at 30 V electric field support these speculations with the temperature reaching $\approx 56^\circ\text{C}$ (Figure 3c-c1) and even higher to about 100°C when the fiber bundle was tied into a knot (Figure 3d-d1).

As a blackbody, the ASF can also respond to irradiation from the surrounding environment and generate heat simultaneously.^[5d] Under simulated solar illumination with an intensity of 1.0 Sun, the photo-to-heat response in the low temperature environment ($\approx 0^\circ\text{C}$) was investigated (Figure 3e,f1). A single ASF woven into a “BIT” letter pattern (Figure 3e1) and network fabric (Figure 3e2) could generate a warm surface at a human comfortable temperature around $19\text{--}21^\circ\text{C}$ under solar irradiation, in contrast to the freezing temperature of the surroundings, indicating a highly efficient photoheat conversion and storage occurred along the single fiber regardless of a cold environment. Furthermore, the temperature of a bundle of the ASF fibers placed directly on the cold surface could reach even higher, $45\text{--}50^\circ\text{C}$ (Figure 3f-f1), more than double compared to the single ASF/cotton fabrics under the same solar irradiation. The temperature–time curves of single ASF and ASF/cotton fabrics were also recorded (Figures S19–S22 and S28, Supporting Information). The fast heating and thermal storage/release (temperature plateau) of the fabrics were demonstrated with an estimated thermal energy conversion and storage efficiency of ASF/cotton Fabric 2, $61.5 \pm 1.6\%$. The fast heat transfer between the ASF fibers is envisaged to contribute to such pronounced and synergistic photothermal response from all the fibers in the bundle/fabrics (Figures S19 and S28).^[15]

To investigate the superhydrophobic surface property of the ASF, static water contact angle measurements were performed at room temperature. As shown in Figure 4a–c and Figure S29 (Supporting Information), the water droplets retained a nearly spherical shape on the surface of a single ASF with an approximate static contact angle of about 158° , confirming the superhydrophobic nature of the fiber surface, while the GAF, graphene/PEG composite fiber, and pure PEG were hydrophobic (124°), hydrophilic ($<60^\circ$), and superhydrophilic (9.4°), respectively. Such a nonwetting property allowed the fiber to be easily removed from the water droplet, which was suspended by a metallic needle, leaving no visible traces of water on the fiber's surface and without the water droplet falling, as shown in Figure 4d. Even though the water droplet was slightly stretched by the fiber just before its departure, the shape of droplet retained, indicating weaker adhesion between the fiber and the water droplet as compared to that between the needle and the droplet.^[16] As expected, the ASF bundle also exhibited the superhydrophobic property, with water droplets dyed in different colors maintained perfectly spherical on the bundle fibers' surfaces (Figure 4e–e1).

The superhydrophobic property renders ASF the ability of waterproof and self-cleaning. After the ASF was immersed in water for 5 min, thermal gravimetric analysis (TGA) weight loss profile shows no trace of water absorption into the fibers and reduction of thermal stability of the fiber, despite the hydrophilic nature of PEG in this case (Figure 4f), mainly owing to FC coating on the surface and two ends of the fibers (Figures S5 and S6 and S30 and S31, Supporting Information). In contrast, the loading content of PEG in GAF without FC coating decreased from ≈ 74 to 66 wt%, with nearly 10% weight loss after soaking (Figure S32, Supporting Information). The

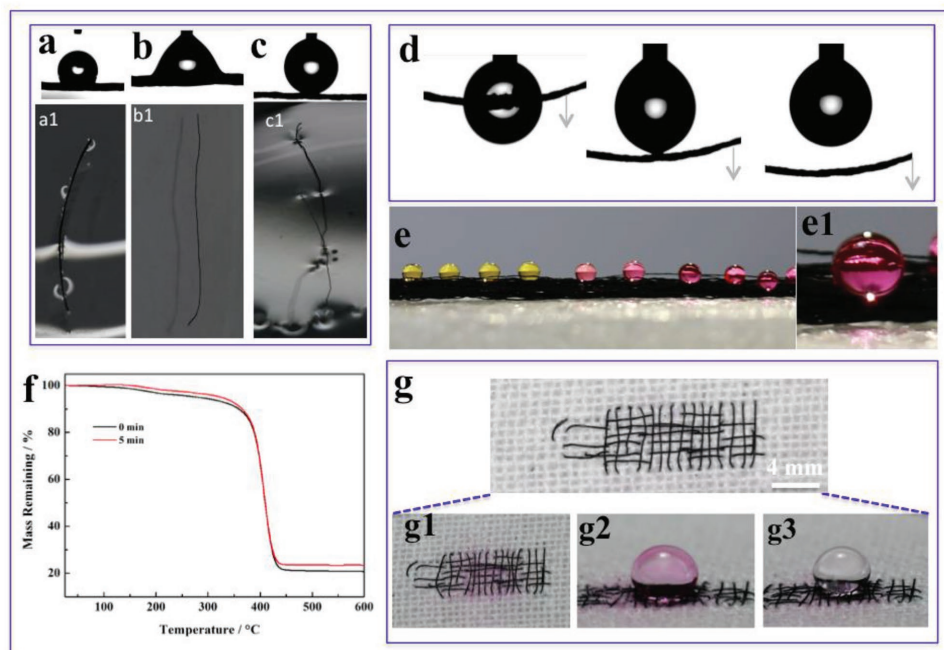


Figure 4. Photographs of the contact angle measurement and graphene aerogel fiber (a–a1), graphene/PEG composite fiber (b–b1), and ASF (c–c1) on the surface of water, pulling out ASF fiber from the water droplet (d). Superhydrophobic surface property of the ASF fiber bundle (e–e1). TGA curves of ASF fiber before and after being soaked in water for 5 min (f). The self-cleaning function of ASF woven fabric (g1–g3).

self-cleaning function was demonstrated on the surface of the ASF woven fabric contaminated by white powders containing rhodamine B (RhB) dye (Figure 4g-g3; Movie S1, Supporting Information): a colorless water droplet became magenta immediately when it contacted the powder on the fiber surface. Subsequently, the magenta water droplet could move around, taking and dissolving the rest of the powder on the superhydrophobic surface of the fiber without changing its shape. Another experiment further demonstrated self-cleaning function of the ASF. When RhB dye solution was dripped on a random ASF coil, the large water droplets could not stay on the mesh instead they went through the fiber mesh and were absorbed by white paper underneath (Figure S33, Supporting Information). As a result, the white paper was dyed magenta while the fiber was not wetted (Figure S33, Supporting Information). Even when the fiber mesh was immersed into magenta water, it still maintained clean without residue dye on the fibers (Figures S34–S36, Supporting Information). The self-cleaning function of the ASF fiber described above reiterates the superhydrophobic nature of the FC-coated fibers. The high contact angle of the ASFs offers low surface energy and weak interfacial tension that prevents water and RhB dye particles from adhering to or diffusing into the fibers.

In brief, a variety of multiresponsive smart fibers with a wide range of tunable phase transition temperatures and enthalpy have been produced through impregnation of different types of organic PCMs into graphene aerogel fiber. The strong capillary action in the highly porous graphene aerogel network generated uniform nanolaminates between graphene sheets and PCM along the microfiber, resulting in overall superior mechanical, electrical, and thermal properties. A diverse range of smart fibers with adjustable phase transition temperatures and phase-change enthalpies have been developed though the selection of different types of PCMs and controlling PCM loading (up to 84 wt%) during the impregnation process with the latent heat reaching up to 124 J g⁻¹ for PEG–ASF, 176 J g⁻¹ for wax–ASF, and 186 J g⁻¹ for C20–ASF. The unique nanostructure and superior properties of PCM nanocomposite fibers have been demonstrated through electrothermal and photothermal functions in response to multistimuli (electron/photon) and environmental irradiation with high efficiency, stability, and reversibility. A superhydrophobic fluorocarbon coating has added a self-cleaning feature and further enhanced the mechanical properties of the fibers. The robustness and compliance of the knot and woven fabric made from these aerogel-directed smart fibers, plus the appealing multiresponsive functions, including energy conversion and storage, is promising for a range of novel smart units for future flexible and wearable devices, e.g., smart wearable fabrics (Figure S28, Supporting Information) and clothes suitable for climbing in frigid zones.

Experimental Section

Materials: Graphite powder (1 μm) was purchased from Qingdao Tianheda Graphite Co., Ltd, Qingdao, China. P₂O₅, K₂S₂O₇, 30% H₂O₂, HCl (aq), and H₂SO₄ were obtained from Sinopharm. Chemical Reagent PEG, *n*-eicosane, and L-AA were obtained from Aladdin Company. Paraffin (OP44E) was obtained from Ruhr Tech, Hangzhou, China.

Fluorocarbon resin was obtained from Sino-Fluorine Science and Technology Ltd, Shenzhen, China.

Preparation of GO-LC: Graphene oxide (GO) sheet was prepared from graphite powder by a modified Hummer method reported in our previous study. Liquid crystalline GO was obtained by the centrifugation of the GO dispersion at high speed (1700 rpm) for 4.5 h.^[5d]

Preparation of Graphene Aerogel Fiber: The resulting GO liquid crystal (≈50 mg mL⁻¹) was spinning into GO/VC (aq) was allowed to stand for 12 h at 50 °C water bath. Followed by washing at least four times with absolute ethyl alcohol to replace the water and supercritical drying with CO₂ (40 °C, 10 MPa) for 12 h, the GAF was obtained.

Preparation of the Aerogel-Directed Smart Fiber: Both GAF and phase-change materials (PEG or paraffin) were heated to 75–80 °C in a vacuum oven and were placed for 3 h for the GAF to be infused with PCM. After the GAF was submerged in the melting PCM, the system was allowed to hang under 80 °C, and then the excess PCM adhered on the fiber surface was removed. After cooling down to room temperature, the composite fiber was obtained. Subsequently coating with fluorocarbon resin, the smart fiber can be obtained. The ASF with different loading PEG was prepared via similar method, and the difference was that the PEG solution was used to replace the melt PEG.

Characterizations: The morphology of the samples was examined by SEM (Hitachi S-4800) with the acceleration voltage of 5–15 kV. The paraffin and the composite were coated with Au nanopowder under current of 20 mA for 2 min. Transmission electron microscope measurement was carried out on Tecnai G2F20 S-Twin with the acceleration voltage of 200 kV; the TEM sample with about 40 nm thickness was prepared by cryocutting; and Raman spectra were recorded on a LabRAM HR Raman spectrometer with a 50 mW He–Ne laser operating at 632 nm with a CCD detector. X-ray diffraction (XRD) patterns were recorded on a D8 Advanced spectrometer with a scanning rate of 0.05 s over an angular range of 5–65° (2θ). The pore size distribution and average pore diameter of the aerogels were analyzed by the BJH nitrogen adsorption and desorption method (ASAP 2020, Micromeritics, USA). The surface area of the aerogels was determined by the Brunauer–Emmett–Teller (BET) method, based on the amount of N₂ adsorbed at pressures 0.05 < P/P₀ < 0.3. Thermal gravimetric analysis and DTG were carried out using a TG 209F1 Libra (NETZSCH) analyzer with a heating rate of 10 °C min⁻¹ in a nitrogen atmosphere. DSC analysis was performed on a DSC 200F3 NETZSCH with a heating and cooling rate of 10 °C min⁻¹. The electric resistances of the aerogel and its composites were measured by using a CH1Chief 600D electrochemical workstation, and the electric conductivity can be calculated by the equation: $\kappa = IL/US$, where κ is the electric conductivity, I is the current that cross the sample, U is the voltage applied in the sample, L is the length of sample current goes through, and S is the cross area of current. The stress–strain curves were measured by an Instron 3365 tensile testing machine. The contact angle was measured by OCA 15EC DataPhysics Instruments GmbH. Infrared photos were taken with a MiniIR (M1100150) camera. The XPS spectra were measured by Escalab 250Xi, Thermo Scientific. The temperature–time curves were measured and recorded by thermal couple and Keysight 34970 Data Acquisition.

Supporting Information

Supporting Information is available from the Wiley Online Library or from the author.

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Conflict of Interest

The authors declare no conflict of interest.

Keywords

electrical/thermal/optical stimuli, graphene aerogels, phase-change materials, smart fibers

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